COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 7

1 Artificial Intelligence (sbh11)

For a two-class classification problem with classes C_1 and C_2, we will use the following linear classifier with probabilistic output

$$\Pr(C_1|\mathbf{x}, \mathbf{w}, w_0, \theta) = \sigma_\theta\left(\mathbf{w}^T\mathbf{x} + w_0\right) \tag{1}$$

where $\mathbf{x}$ is an input vector and $\mathbf{w}$, w_0 and θ are parameters, and

$$\sigma_\theta(x) = \frac{1}{1 + \exp(-\theta x)}$$

is the activation function.

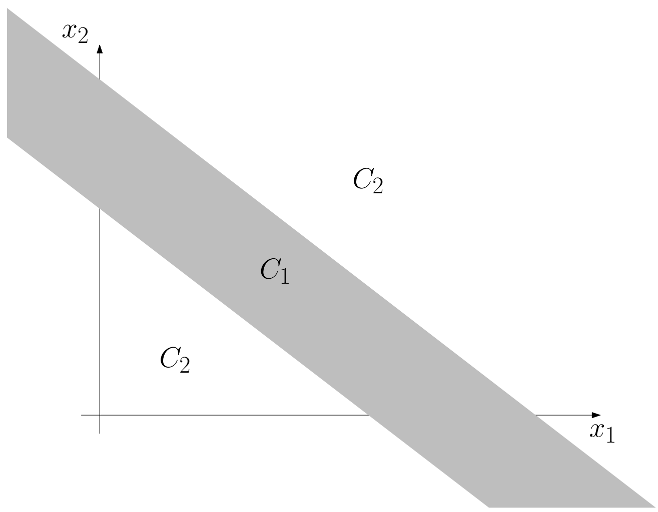
Linear classifier

(*a*) The activation function σ_θ often appears assuming $\theta = 1$. What role does θ play when its value can be set freely? [1 mark]

Answer: It sets the rate at which the output of the classifier changes from 0 to 1—small values lead to a slow transition and large values to a fast one.

Activation function

(*b*) You are provided with a collection $\mathbf{s} = ((\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_m, y_m))$ of m training examples where $y_i \in \{0, 1\}$ for $i = 1, \ldots, m$. You suspect that the data has a band-like structure. In two dimensions:



Explain how the linear classifier can be modified, changing only its activation function and without adding further parameters, such that it can be trained on this type of data. It should be possible to set the width of the C_1 region as part of the training. [6 marks]

Answer: The σ_θ function moves from 0 to 1 smoothly, passing through 0.5 when its argument is 0. We need something that moves from 1, down to 0 for an argument of 0, and back to 1 (1 mark). The function

$$\sigma'_\theta(x) = 4\left(\sigma_\theta(x) - \frac{1}{2}\right)^2$$

has exactly the shape needed (3 marks). It also provides for θ to be used to set the width (2 mark). Other functions also suffice.

Gradient descent

(*c*) When using a two-class classifier $h(\mathbf{x}, \mathbf{p})$ that has parameters $\mathbf{p}$ and outputs probabilities, one often uses the error measure

$$E(\mathbf{p}) = -\sum_{i=1}^{m} y_i \log h(\mathbf{x}_i, \mathbf{p}) + (1 - y_i) \log(1 - h(\mathbf{x}_i, \mathbf{p})).$$

Explain how the gradient descent method can be applied to solve this classification problem for the data described in Part (*b*), giving explicit formulae for the gradients. [13 marks]

Answer: This can be achieved using straightforward gradient descent, for which the definition should be given (1 mark). We therefore need to derive the gradient, which means deriving the derivatives of $E(\mathbf{p})$ with respect to the parameters (1 mark). This can be done as follows, noting that $\mathbf{p} = [\mathbf{w}, w_0, \theta]$ and $h(\mathbf{x}_i, \mathbf{p}) = \sigma'_\theta \left(\mathbf{w}^T \mathbf{x}_i + w_0\right)$, and denoting any of the available parameters by λ, we have

$$\frac{\partial E}{\partial \lambda} = \sum_i y_i \frac{1}{\sigma'_\theta(\cdot)} \frac{\partial}{\partial \lambda} \sigma'_\theta(\cdot) - (1 - y_i) \frac{1}{1 - \sigma'_\theta(\cdot)} \frac{\partial}{\partial \lambda} \sigma'_\theta(\cdot)$$

where $\mathbf{w}^T \mathbf{x}_i + w_0$ appears as $\cdot$ in the above equation (2 marks). It is therefore only necessary to compute the derivative of $\sigma'_\theta(\cdot)$ with respect to each parameter. This is straightforward:

$$\frac{\partial}{\partial \theta} \sigma'_\theta(\cdot) = 4 \frac{\partial}{\partial \theta} \left(\sigma_\theta(\cdot) - \frac{1}{2}\right)^2 \tag{2}$$
$$= 8 \left(\sigma_\theta(\cdot) - \frac{1}{2}\right) \frac{\partial}{\partial \theta} \sigma_\theta(\cdot) \tag{3}$$
$$= 8 \left(\sigma_\theta(\cdot) - \frac{1}{2}\right) \frac{(\cdot) \exp(-\theta(\cdot))}{(1 + \exp(-\theta(\cdot)))^2} \tag{4}$$

relying on the fact that $\cdot$ does not depend on θ (4 marks).

$$\frac{\partial}{\partial w_0} \sigma'_\theta(\cdot) = 4 \frac{\partial}{\partial w_0} \left(\sigma_\theta(\cdot) - \frac{1}{2}\right)^2 \tag{5}$$
$$= 8 \left(\sigma_\theta(\cdot) - \frac{1}{2}\right) \frac{\partial}{\partial w_0} \sigma_\theta(\cdot) \tag{6}$$
$$= 8\theta \left(\sigma_\theta(\cdot) - \frac{1}{2}\right) \sigma_\theta(\cdot)(1 - \sigma_\theta(\cdot)) \frac{\partial}{\partial w_0} (\mathbf{w}^T \mathbf{x}_i + w_0) \tag{7}$$
$$= 8\theta \left(\sigma_\theta(\cdot) - \frac{1}{2}\right) \sigma_\theta(\cdot)(1 - \sigma_\theta(\cdot)). \tag{8}$$

This uses the identity $\partial \sigma_\theta(x)/\partial x = \theta \sigma(x)(1 - \sigma(x))$ (4 marks). The derivative $\partial/\partial w_j$ for any j is exactly the same as for $\partial/\partial w_0$ until the last line, where

$$\frac{\partial}{\partial w_j} (\mathbf{w}^T \mathbf{x}_i + w_0) = \mathbf{x}_i^j.$$

(1 mark).
